IGME 106 Group Game Project
Spring 2024

Team: **Hindenburg**

Game: Sticky Situation

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Team and Production

Student Information

Team Name: Hindenburg

Team Members:

Name	*Email*	*GitHub username*	*Area(s) of Expertise*
Dalton Zeigler	drz5712@g.rit.edu	Dalton-Zeigler	Debugging, Organization, Commenting
Jon Rysdam	jer7616@g.rit.edu	WalrusesAreAwesome	Non-object oriented programming, algorithms, vectors
Kevin Zheng	kz3175@g.rit.edu	NadalsAvailable	Debugging, File IO, Commenting, OOP, Art
Thor Frostick	tdf4115@g.rit.edu	ThorFrostick	User input, commenting, vector math
Zachary Stulberg	zs3818@rit.edu	Zasnyy	Programming/writing code, Code Design/Organization, Math, Debugging

Game Repo and Presentation Slides

GitHub repo link:

Task board link: https://trello.com/b/9H34Ss3q/hindenburg-timeline

Links to Google Slides:

Milestone	*Google Slides*
1 - Team Formation & Conceptualization	Hindenburg Formation & Concept

2 - Game Loop	Hindenburg - Sticky Stiuation - Milestone 2
3 - Data-Driven Prototype	Hindenburg Milestone 3 Slideshow
4 - Polish	Milestone 4 Slideshow

Team Goals and Mission Statement

Discuss each of your individual and team goals for this project (beyond “get a good grade and pass the class”).

Individual Goals:

Team member name:	*Their individual goals:*
Dalton Zeigler	To have a basic understanding of developing a game in a group and by myself. To also have a basic grasp of C# coding.
Jon Rysdam	I’d like to make a new “best game in my portfolio” in my own opinion. Especially now with an “actual” game engine.
Kevin Zheng	I want to make a functional prototype of a game and have some experience of team working.
Thor Frostick	I want to make a working prototype of a game that is able to function properly without any major errors or glitches. I also wish to hone my programming skills and learn how exactly to make a game of my own.
Zachary Stulberg	I want to use all the skills and knowledge that I have acquired to finally put together a fully-fledged game that I can be proud of.

Team Goals:

Goal #1: Complete designing a game without any major internal issues.

Goal #2: Do something in this game we haven’t done before.

Goal #3: To have a better understanding of designing a game from the ground up.

Decide on an overall team mission statement.

For example, do you want a strong portfolio piece? Learn new technical skills? Improve project management skills? What are your priorities? Having (and documenting!) this conversation early will help keep everyone moving in the same direction when the project gets busier later in the semester.

Create a functioning game prototype that can be used as a portfolio piece that shows we know how to work together in a team and design a game from the ground up.

Group Norms

List your group's norms here:

- Each member dedicates a reasonable time to contribute to completing the game, currently agreed upon 2-4 hours a week for each member
- Each member shall contribute to meetings, with the minimum requirement being that we each discuss what we have done and what needs to be accomplished.
- To tell everyone about changes you have done to the game with your portion or someone else's

Milestone 1: Design

Design

Give a brief description of your game:

Our concept is a platformer where you play as a slime-like creature that can attach to walls and swing on them in order to reach the end of the level. The game will incentivise getting a faster time and giving a better score for quicker completions.

Answer the following questions:

What genre is your game?

2D fast paced platformer

What are the objectives?

Beat the levels, get better times

How many players?

Singleplayer (1)

What does the player control (weapon, puzzle blocks, etc.)? What is their goal and motivation? Why do they want to keep playing?

The player will control a grapple-like object like Spiderman Web-Slinger.
Hopefully the player will continue playing in order to achieve faster times. Getting a certain time will give you a higher rank.

How does the player progress through the game? What is the win condition?

The player will be able to slowly move along the ground and walls with input, but they will be incentivised to jump, as there will be a swinging mechanic where the player can shoot out a Spiderman-like web to swing through obstacles. The goal is to reach the end of the level, which will have an object to signified

Are there NPC's in the game? If so, how does the player interact with them? What is the result of those interactions?

There will be a moving enemy (possibly a spike ball) that, when the player touches them, will cancel a swing and/or knock back the player, stunning them for a short time.

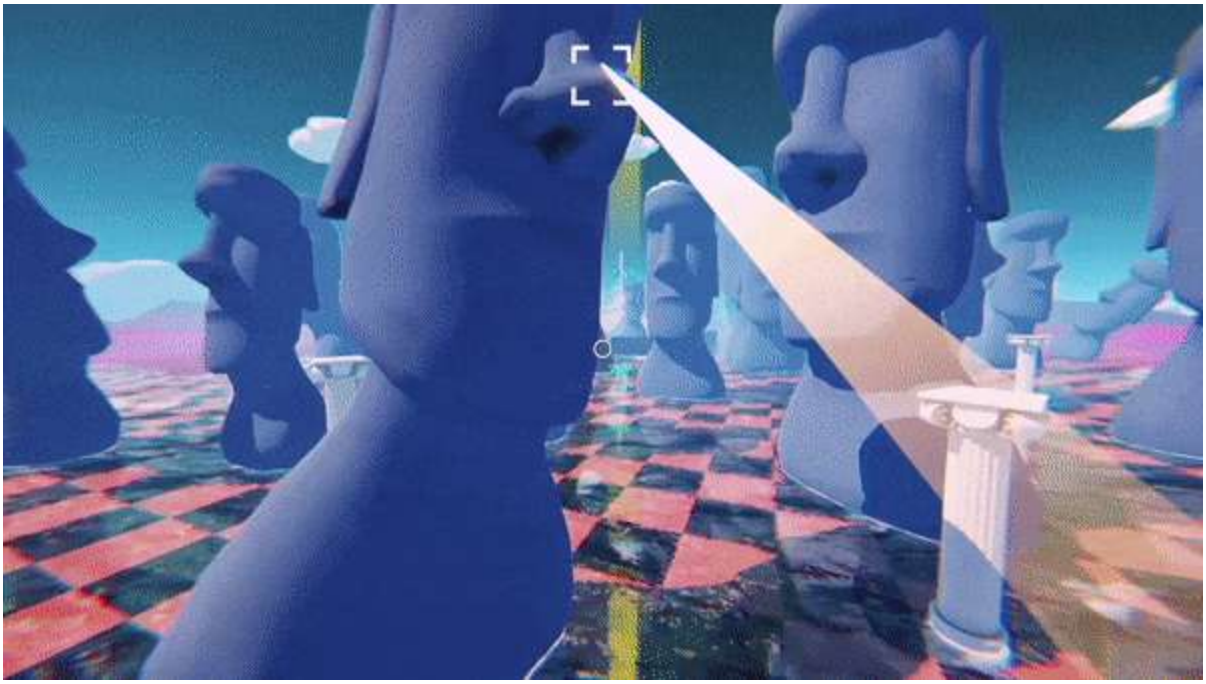
What is the "control scheme" - what controls will the player utilize?

Movement along the ground or walls will use the A and D keys left and right respectively, and will be able to jump with the Space key.

The grapple controls are WS when the grapple is attached to the wall or object to increase or decrease the length of the Grapple.

Did you draw inspiration from any games? If so, how is your game different?

This game draws a lot of inspiration from Verlet Swing (which draws inspiration from Spiderman). In both games, you navigate a world by swinging from a rope extending from your body. This game is different from those in that it is 2D, it has a different theming, and will most likely have different sub-mechanics as a result. It also draws speedrunning inspiration from Celeste.



Whoosh!

Who is your intended audience?
People who like fast paced games, or platforming games. We believe this game would be rated E for everyone.

Why is your game fun?
It will be fun to move quickly with the swinging mechanic, and will have a good amount of replayability as the players go for faster times.

How does it fit within the theme of either *"Just a stick"* or *"The only thing they fear is you"*?
The slime the player controls can stick to walls and platforms, which allows them to land on the underside of platforms.

Scope

Describe your group's overall priorities and rationale that will guide your development priorities. *For example, for your game, which is critical: prototyping a variety of NPC's or showcasing one NPC that's done really well? Are several levels a key feature of your game, or is one superb level ideal?*

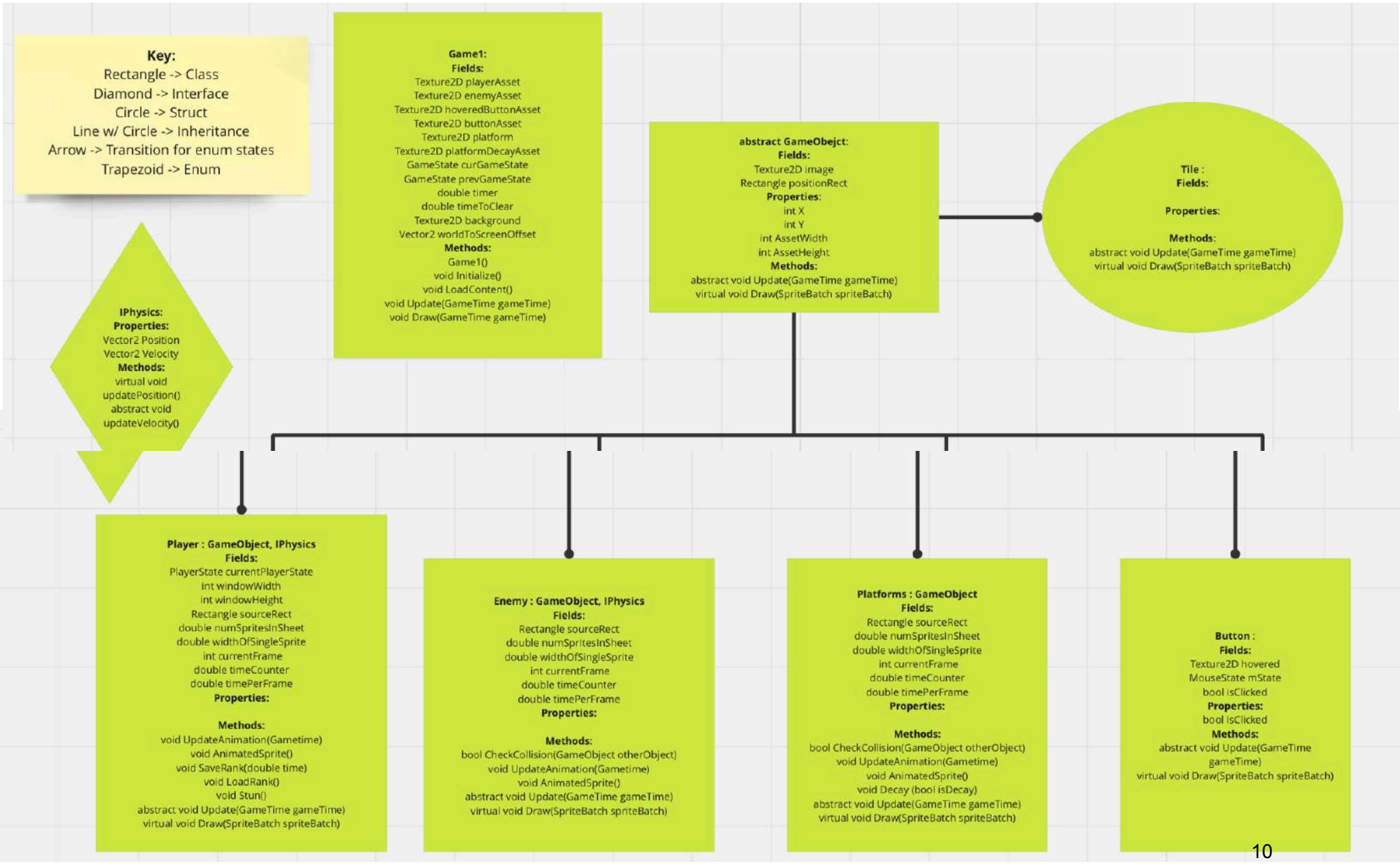
For the physics of sticking to surfaces to function properly

Our focus is currently on designing a single well made level, as well as one specific type of enemy, a rolling spiked ball, that has specific behaviors, such as rolling from the top of the screen on one side to the bottom of the other and knocking back the player if it hits them.

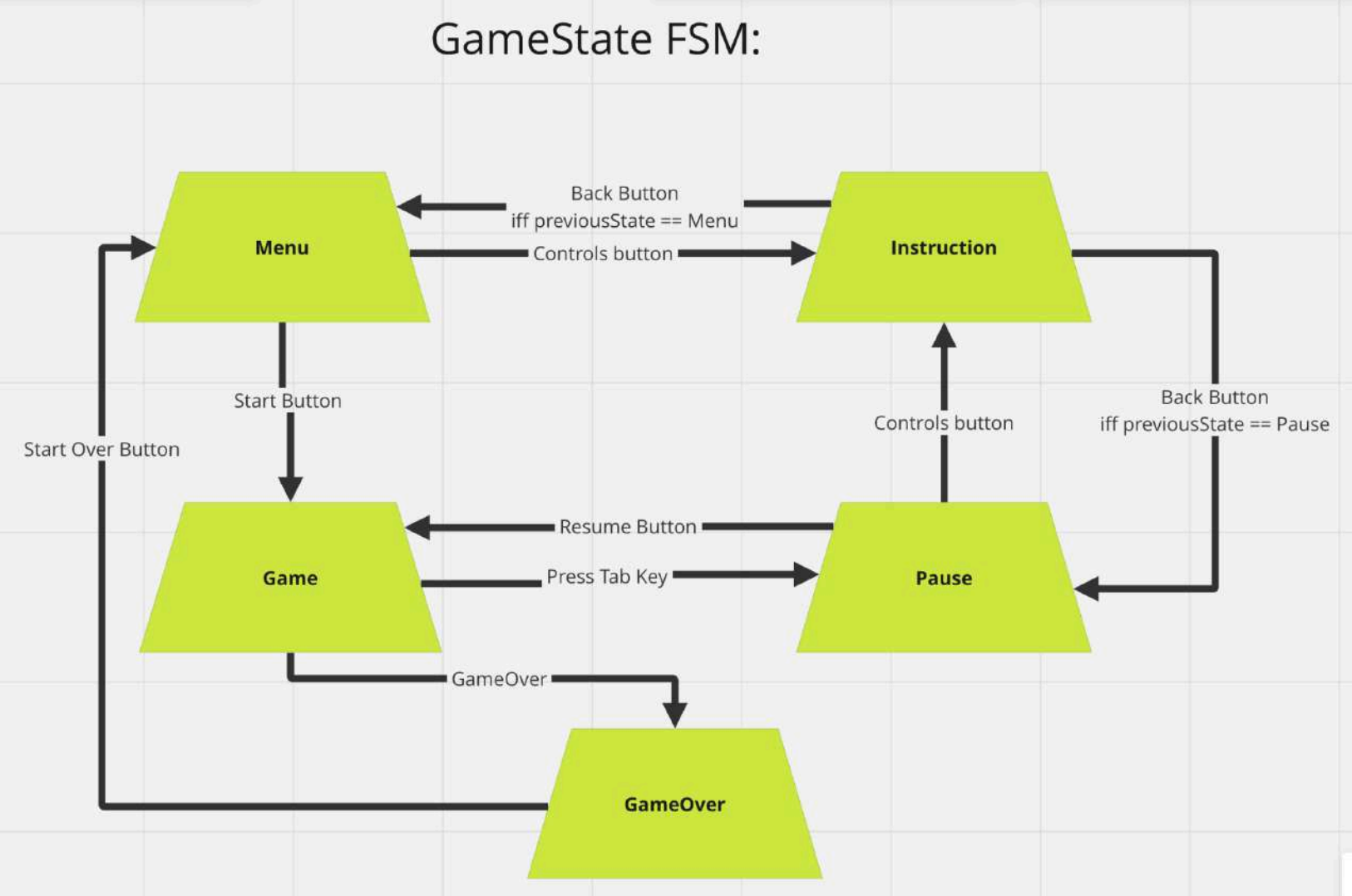
Architecture

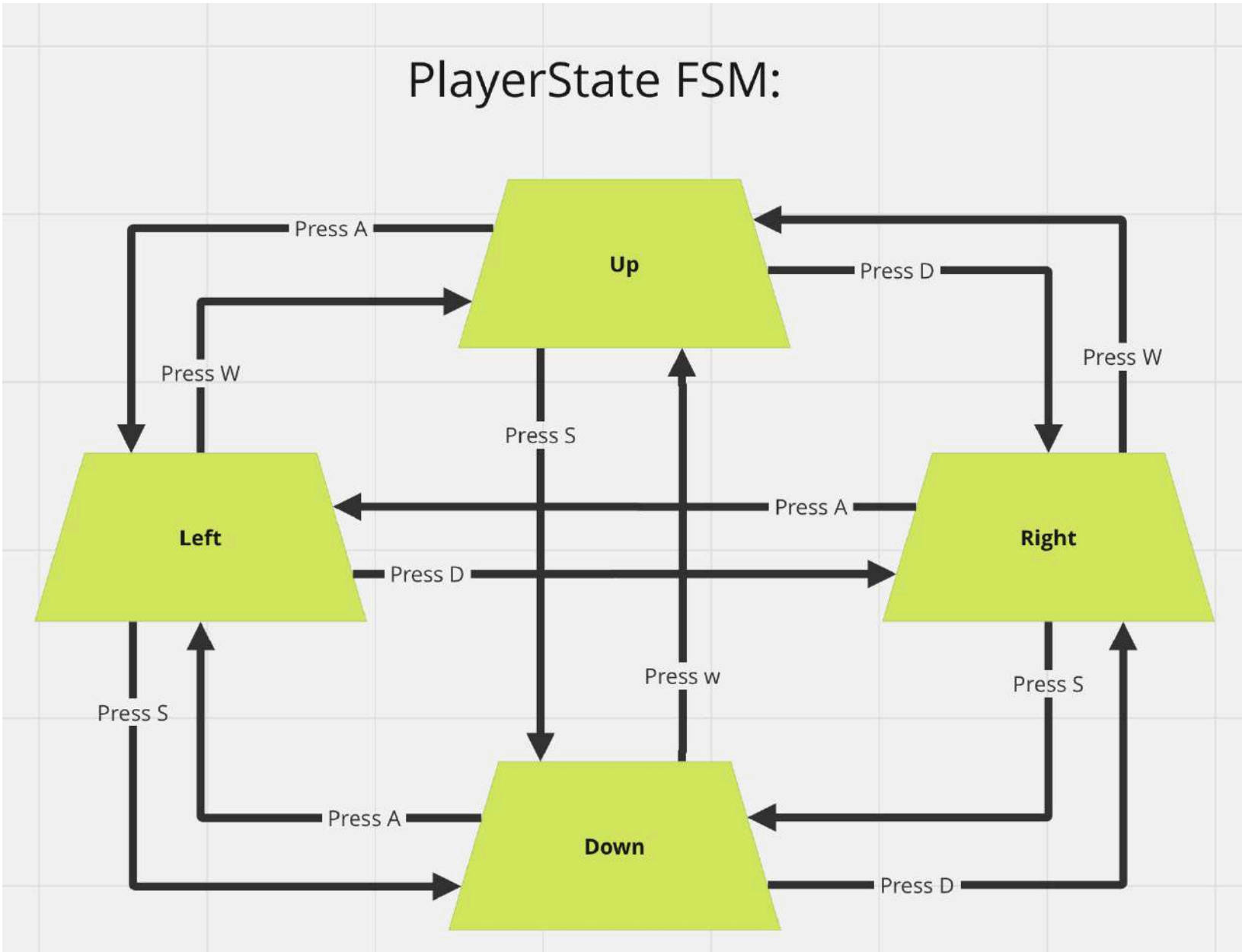
Include your architecture diagram(s) here.
If any classes are abstract, notate that on your diagram.
If there are any interfaces, notate that on your diagram, too.
(Note: If diagrams are large and unreadable, please blow up the image and break it into separate images.)

Game diagram
Link to Miro:
https://miro.com/welcomeonboard/Tm1pVXBZUndIUTZwdkdYOGJRcXRIOW1sSmNRdIljTVIUSWJVcFg3ZmNRS1RiQnISOHIyOTVkbUxoSINEVTIyMnwzNDU4NzY0NTM0ODU0NDY1NzcyfDI=?share_link_id=122071471210


Key:
Rectangle -> Class
Diamond -> Interface
Circle -> Struct
Line w/ Circle -> Inheritance
Arrow -> Transition for enum states
Trapezoid -> Enum
Game1:
Fields:
Texture2D playerAsset
Texture2D enemyAsset
Texture2D hoveredButtonAsset
Texture2D buttonAsset
Texture2D platform
Texture2D platformDecayAsset
GameState curGameState
GameState prevGameState
double timer
double timeToClear
Texture2D background
Vector2 worldToScreenOffset
Methods:
Game1()
void Initialize()
void LoadContent()
void Update(GameTime gameTime)
void Draw(GameTime gameTime)
abstract GameObejct:
Fields:
Texture2D image
Rectangle positionRect
Properties:
int X
int Y
int AssetWidth
int AssetHeight
Methods:
abstract void Update(GameTime gameTime)
virtual void Draw(SpriteBatch spriteBatch)
Tile :
Fields:
Properties:
Methods:
abstract void Update(GameTime gameTime)
virtual void Draw(SpriteBatch spriteBatch)
IPhysics:
Properties:
Vector2 Position
Vector2 Velocity
Methods:
virtual void updatePosition()
abstract void updateVelocity()
Player : GameObject, IPhysics
Fields:
PlayerState currentPlayerState
int windowWidth
int windowHeight
Rectangle sourceRect
double numSpritesInSheet
double widthOfSingleSprite
int currentFrame
double timeCounter
double timePerFrame
Properties:
Methods:
void UpdateAnimation(Gametime)
void AnimatedSprite()
void SaveRank(double time)
void LoadRank()
void Stun()
abstract void Update(GameTime gameTime)
virtual void Draw(SpriteBatch spriteBatch)
Enemy : GameObject, IPhysics
Fields:
Rectangle sourceRect
double numSpritesInSheet
double widthOfSingleSprite
int currentFrame
double timeCounter
double timePerFrame
Properties:
Methods:
bool CheckCollision(GameObject otherObject)
void UpdateAnimation(Gametime)
void AnimatedSprite()
abstract void Update(GameTime gameTime)
virtual void Draw(SpriteBatch spriteBatch)
Platforms : GameObject
Fields:
Rectangle sourceRect
double numSpritesInSheet
double widthOfSingleSprite
int currentFrame
double timeCounter
double timePerFrame
Properties:
Methods:
bool CheckCollision(GameObject otherObject)
void UpdateAnimation(Gametime)
void AnimatedSprite()
void Decay (bool isDecay)
abstract void Update(GameTime gameTime)
virtual void Draw(SpriteBatch spriteBatch)
Button :
Fields:
Texture2D hovered
MouseState mState
bool isClicked
Properties:
bool IsClicked
Methods:
abstract void Update(GameTime gameTime)
virtual void Draw(SpriteBatch spriteBatch)

Include a finite state machine diagram of your game's states:




PlayerState FSM:
Up
Left
Right
Down
Press A
Press D
Press W
Press W
Press S
Press A
Press D
Press w
Press S
Press S
Press A
Press D

Describe which classes and features are data-driven. What will the file format be for those classes and features? *(File format means the type of data and structure of the data found in a file. For instance, a file describing enemy information could be serialized to include the enemy's name, level, starting health, preferred weapon, damage, list of objects in its inventory system, etc.)*

Our plan is to write down a list of high scores as a leaderboard onto a file. This file will be read from in order to compile the leaderboard every time the game is run. For instance, each line contains rank, score, and time (rank|score|time).

Our background class is responsible for tile management, which is going to use file IO for tiling.

Our level data is going to use file IO for positioning, textures, and size based on how big the assets are.

Which classes handle multiple game entities (characters, enemies, NPC's, etc.)?

The Player objects/class, Enemy objects/class, Special Platforms, and Menu Buttons are all inheriting from the abstract GameObjects class, which also is being inherited from by the class for regular tiles.

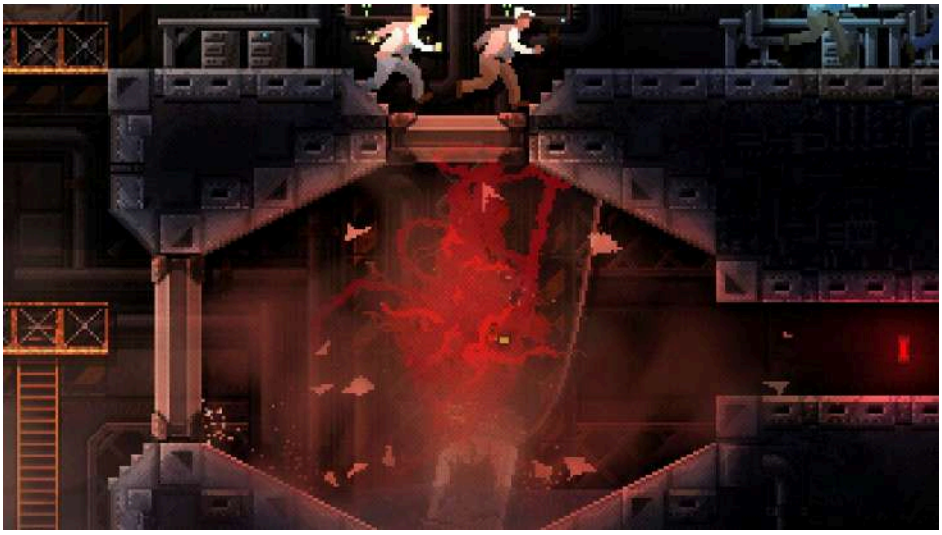
Where and how is player input handled?

For movement, we have a finite state machine that will control the direction the player moves. Additionally, we will check if the mouse is clicked to determine if the player will start swinging. Moreover, we have a separate finite state machine to determine the current state of the game based on the player's input.

Art

What is the game's visual style? Describe what the player will see.
Cartoony style pixel art

Include any concept art for reference. This could be screenshots of current games you've played and are taking inspiration from, a mood board, or custom drawings from team members. Include those references here. If your team already has the assets you will use, include those here instead of concept art.



Less blood and dismay in actual game

Fewer spikes and diamond dealy-bobs

Gray and red instead of gray and green.

Where will you gather art? Include links to potential 2D assets here:
(You don't have to have your full set of art yet, but having a set of images that are potential contenders for your game is good planning. Find them now, not later.)

https://itch.io/game-assets/free

Slime: https://rauszy.itch.io/slime

Interface

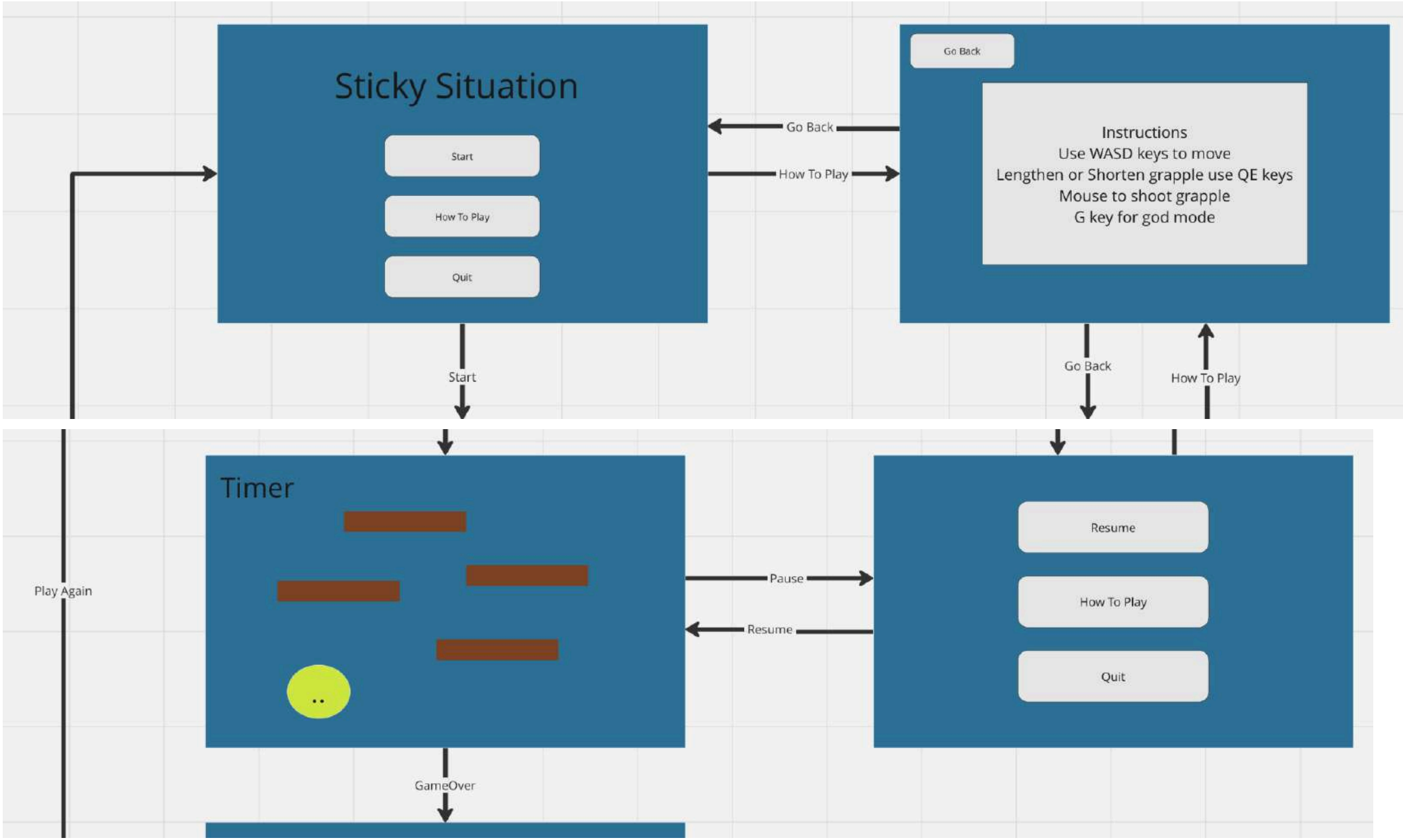
How does the user interact with your game? (*Are you using both keyboard and mouse, or just one of those?)*

In the game state, we will have the player utilize the keyboard to move left and right, the space bar will be used for jumping, and the mouse will be used to aim, fire, and release the sticking arm for swinging the player.

What are the user interface requirements? *(What buttons, bars, UI elements does your game need?)*

On the start screen, we will have a title, as well as Start, How to Play, Leaderboard, and Quit buttons that the player can click on. In the game, the screen will have movement keys on the bottom of the screen, and there will be a menu button shown in the top left corner that the player can click on.

Include mockups of your game screens/menus/HUD here:
(These can be hand-drawn or digitally created.)


Sticky Situation
Start
How To Play
Quit
Go Back
How To Play
Go Back
Instructions
Use WASD keys to move
Lengthen or Shorten grapple use QE keys
Mouse to shoot grapple
G key for god mode
Start
Go Back
How To Play
Play Again
Timer
Pause
Resume
Resume
How To Play
Quit
GameOver

GameOver
Score
Rank here
Play Again
Quit

Production

When and where does your group meet? Who is the "record-keeper" for meetings? *(Does the record keeper rotate, or is it always the same person?)*

We decided to meet Saturdays from 1pm-3pm, Sundays from 1pm-4pm, and potentially Tuesdays from 5:30-7:30. As for location, we decided to meet up at different the Commons restaurant on Saturdays, the Golisano computer labs on Sundays, and we will decide where to meet on Tuesdays if we have to. We also decided that Thor Frostick will maintain records for the meetings.

What is your group's chosen method of communication?

We created a Discord server to maintain contact outside of classes and meetings, and we communicate face to face in classes and meetings.

How often are group members expected to communicate with each other?

We are expecting each other to relay to everybody if some issue comes up with the project or they can't make it to meetings. Additionally, we believe that communicating regularly over the week is important, though we won't necessarily require daily communication.

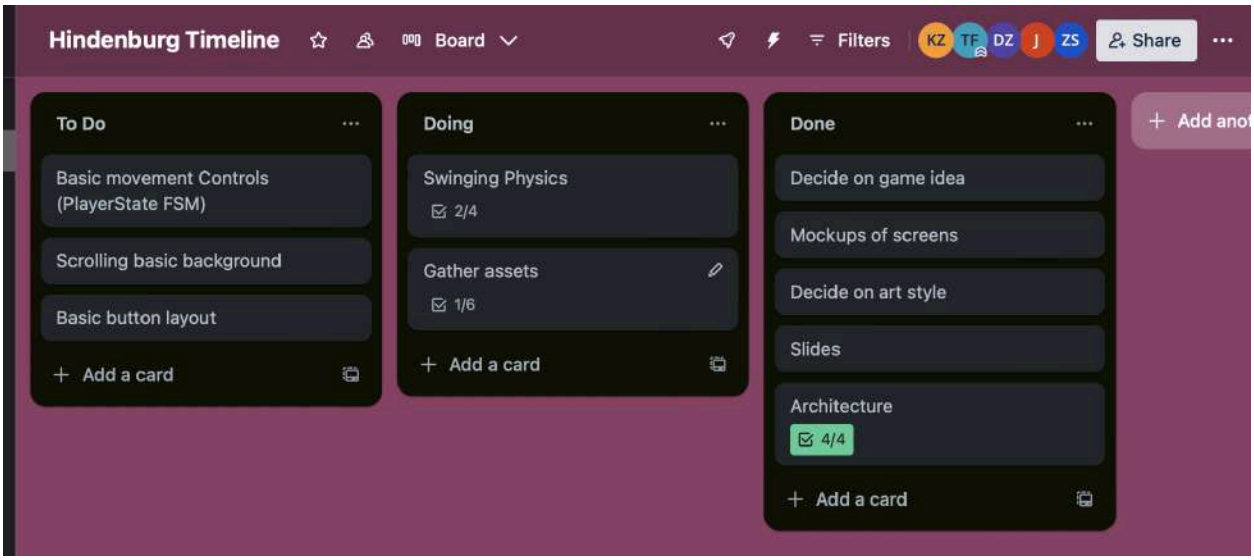
How will your group utilize Trello (or another Kanban board) to handle task management?

We will use a Trello board to organize what needs to be done, what people are currently working on, and what has been completed.

How will your group assign/divide programming responsibilities throughout the project?

Each member will be responsible for specific tasks, which will be decided upon as a team. We will most likely decide at the beginning of meetings who will work on what tasks, though there are certain tasks that individual members of our group are better suited for.

Include a screenshot(s) of your Kanban board by the end of this milestone:



Game Requirements

What are the minimal specifications you need to have a working version of the game by the end of the semester? Include a brief description of those requirements.
(For example, "1 level with 3 enemy types and 2 power-ups. Background tiling system for the single level. Art assets for the player and each type of enemy," et cetera)

We are requiring a functioning playable slime character with a proper swinging mechanic, a single finished level, and a certain number of enemies with consistent movement.

What are your stretch goals for the game?

We would like to try and include more levels if we have the time for such an addition.

Task Timeline for Milestone 2

Include a bulleted list of tasks to complete during milestone 2:

- Basic movement Controls

- Scrolling Basic Background
- Basic Button layout

Lessons Learned

Write a 2 - 3 paragraph summary of lessons learned (both good and bad!) during this milestone.
This should be done after a group discussion of your group processes.

We started off our group project a little rocky with some hiccups. Firstly, we took a bit longer than perhaps expected to begin working on the architecture of the game, and had to quickly determine what classes and methods our project might need. Additionally, because we spent as much time on the architecture as we did, it caused the pace of our work on the slides to falter, and we did not complete them until the necessary due date. Furthermore, while we did set up a good communication system and meeting schedule, we are considering lowering the amount of weekly meetings for the next milestone, since it is mostly coding.

However, that is not to say our project was ineffective. On the contrary, we managed to establish many good habits to our team, as well as the groundwork for our game. When we first met up we quickly established what the meeting times were going to be. We came up with our theme to "Just a Stick" rather quickly. On the first group work day we came up with the theme of escaping an abandoned lab. On that same day we also laid the groundwork of the mechanics of our game. Then on the 17th to 18th of February we started and finished the design of the architecture. The design of our architecture went pretty smoothly.

Describe how your experience this milestone will influence your plans (workflow, communication, task management, etc.) for the next milestone.
Again, done in tandem with your group. What will you keep doing well? What changes will your group make?

We will not push the work to the last few days. Our meetings began to stall, even in this more creative-intensive portion of the process, so we will probably meet for less time in the future. We believe that we had a decent work pace once we began working, and it would be most effective to continue that pace.

Milestone 2: Bare-Bones Game

Link to your group's presentation slides:

Hindenburg - Sticky Stiuation - Milestone 2

Scope

"MVP" means "minimum viable product" -- the bare minimum functionality that needs to be complete in order for the game to ship. For this class, "ship" means it demonstrates your overall priorities and meets the course requirements.

Given the priorities, what is your target MVP? Why?
(This is NOT just a list of features. Describe the target player experience of your prototype!)

Our goal is to have a single well-made level where the player can navigate past enemies with set movement using the slingshot mechanic to shoot off from platforms and walls. They will ascend up through the level to reach the goal, at which point a score is given based on the time it took to complete, with a rank given based on that time. (i.e, S <= 1:30 minutes, A <= 2:00 minutes, B <= 2:30 minutes, etc)

Team Responsibilities

List each team member's responsibilities during this milestone.

Team Member:	Responsibilities:
Thor Frostick	Enumerations, FSM's,
Jon Rysdam	Physics
Zach Stulberg	Player and Enemy objects, Game Assets
Kevin Zheng	Backgrounds, Level Design
Dalton Zeigler	Buttons, Drawing

Production

How has your group utilized the task management strategy(ies) that you chose in the first milestone?

We used a Trello board to keep track of what we had done and what we needed to do and specifics of each task. In addition, we also communicated with each other using discord, notifying each other what changes have been made to the code.

Challenges

What challenges has your team faced with the project? How has your team overcome those challenges?

We were having difficulties with implementing the grappling mechanic and scrolling. As for the first challenge, we realized that with the given time we have, we will not be able to implement the grappling mechanic in the game, so we decided to replace it with a slingshot mechanic. And for the second challenge, we tried to look it up online, but that did not work, so we went to the tutoring center and the professor for assistance.

Are there bugs that require fixing? If so:
What are they?

Logic errors with tiling and button functionalities.

What is your team's plan for tracking and then fixing the bugs?

Note these bugs on the Trello board and fix them during milestone three (as early as possible/preferably the first week of milestone three).

Design/Architecture/Style/UI/Production Changes

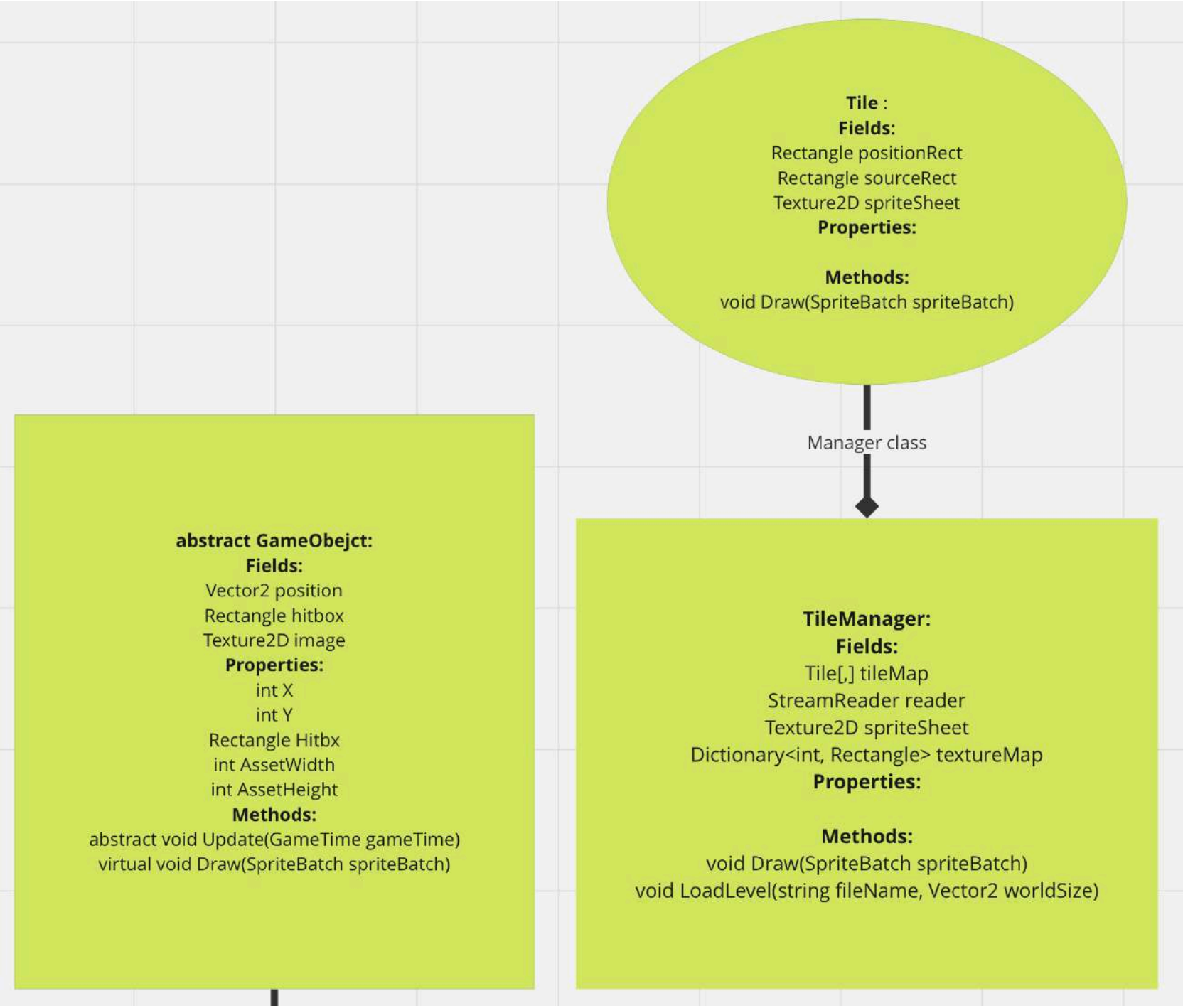
If your design has changed, describe the changes here:

We replaced the grapple mechanic with a slingshot mechanic. Instead of grappling, the player can "shoot" themselves (the slime character) around the screen with mouse clicks.

If your architecture has changed, include an updated architecture diagram here:

Key:
Rectangle -> Class
Diamond -> Interface
Circle -> Struct
Line w/ Circle -> Inheritance
Arrow -> Transition for enum states
Trapezoid -> Enum

Game1:
Fields:
bool gameFinished
TileManager tileManager
Texture2D tileSpriteSheet
Vector2 velocity
Texture2D playerImg
Player slime
float scale
float speed
Vector2 playerWorldCenter
Texture2D worldImg
Background background
Vector2 screenSize
SpriteFont buttonFont
List<Button> startingMenuButtons
List<Button> pauseButtons
List<Button> gameOverButtons
List<Button> howToPlayButtons
KeyboardState prevKBState
KeyboardState curKBState
Helper Methods:
bool SingleKeyPress(Keys key, KeyboardState curState, KeyboardState prevState)
void LeftClickCount()
void StartGame()
void ResumeGame()
void Instructions()
void Back()
void StartAgain()
void Quit()


Tile :
Fields:
Rectangle positionRect
Rectangle sourceRect
Texture2D spriteSheet
Properties:
Methods:
void Draw(SpriteBatch spriteBatch)
Manager class
abstract GameObejct:
Fields:
Vector2 position
Rectangle hitbox
Texture2D image
Properties:
int X
int Y
Rectangle Hitbx
int AssetWidth
int AssetHeight
Methods:
abstract void Update(GameTime gameTime)
virtual void Draw(SpriteBatch spriteBatch)
TileManager:
Fields:
Tile[,] tileMap
StreamReader reader
Texture2D spriteSheet
Dictionary<int, Rectangle> textureMap
Properties:
Methods:
void Draw(SpriteBatch spriteBatch)
void LoadLevel(string fileName, Vector2 worldSize)

nent()

Player : GameObject, IPhysics
Fields:
PlayerState currentPlayerState
Vector2 worldPos
Rectangle sourceRect
Vector2 velocity
double timeCounter
float scale
float speed
Vector2 screenSize
Properties:
Vector2 Position
Vector2 Velocity
int Width
int Height
Methods:
override void Update(GameTime gameTime)
void Draw(SpriteBatch spriteBatch, Vector2 worldScreenOffset)
public void MovementUpdate()
public void SaveRank(double time, string fileName)
void LoadRank()
void Stun()

Enemy : GameObject, IPhysics
Fields:
Vector2 worldPos
Rectangle sourceRect
Vector2 velocity
float scale
float speed
Vector2 screenSize
Properties:
Vector2 Position
Vector2 Velocity
int Width
int Height
Methods:
override void Update(GameTime gameTime)
void Draw(SpriteBatch spriteBatch, Vector2 worldScreenOffset)
void MovementUpdate()
bool CheckCollision(GameObject otherObject)

delegate void OneButtonClickDelegate()
Button :
Fields:
SpriteFont font
Rectangle rect
Vector2 textLoc
string text
Texture2D buttonImg
Color textColor
MouseState prevMState
event OneButtonClickDelegate OnLeftButtonClick
Properties:

Methods:
void Update()
void Draw(SpriteBatch spriteBatch)

IPhysics:
Properties:
Vector2 Position
Vector2 Velocity
Methods:
abstract void UpdateMovement()

Pla

Pl

Background:

Fields:

Vector2 worldScreenOffset

Vector2 worldSize

Texture2D worldImg

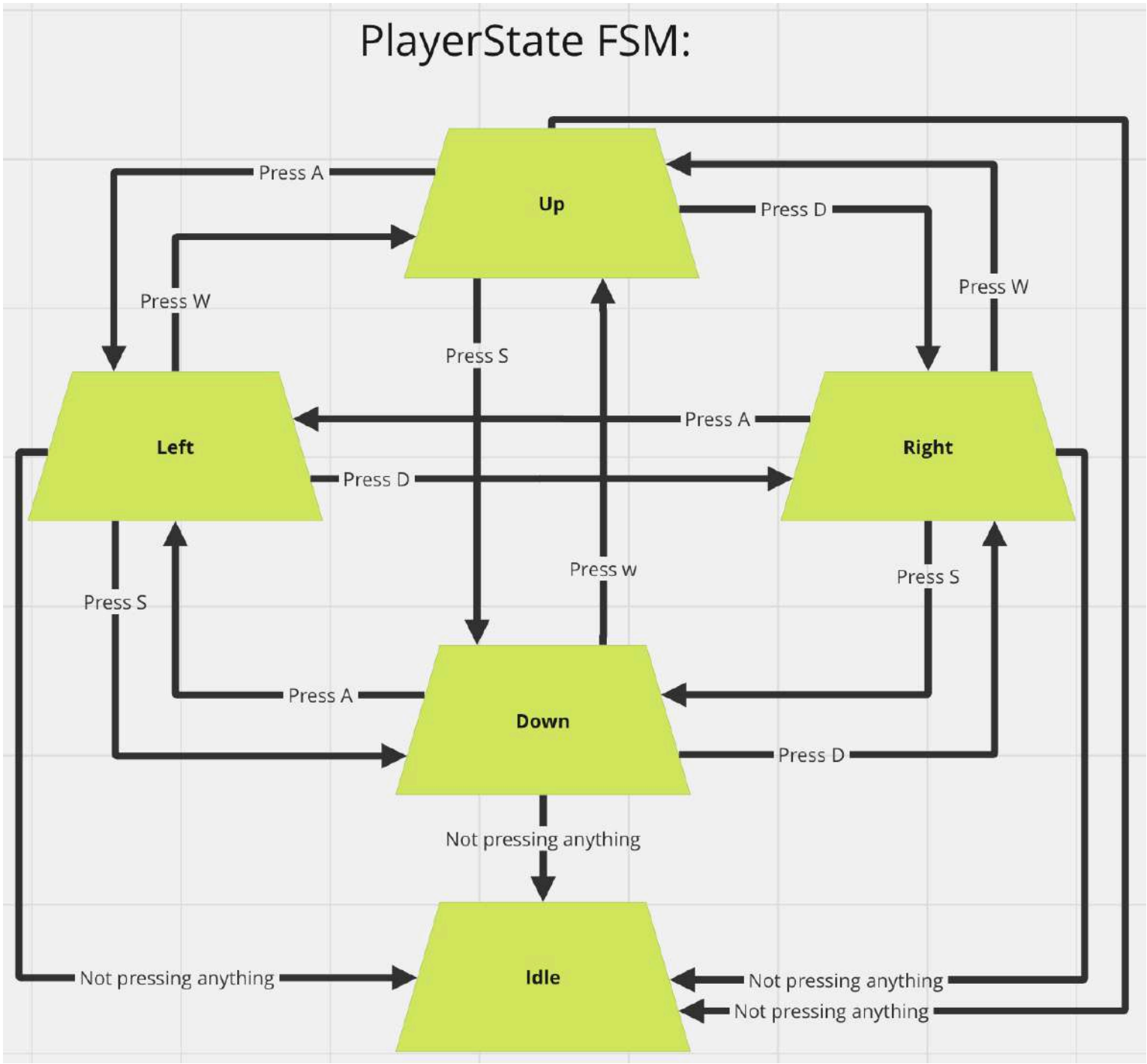
Properties:

Vector2 WorldScreenOffset

Methods:

void Update(GameTime gameTime, Vector2 screenSize, Player player)

void Draw(SpriteBatch spriteBatch)


PlayerState FSM:
Up
Left
Right
Down
Idle
Press A
Press D
Press W
Press W
Press S
Press A
Press D
Press S
Press w
Press S
Press A
Press D
Not pressing anything
Not pressing anything
Not pressing anything
Not pressing anything

If your visual style has changed, describe the changes here and include new reference images:

No changes.

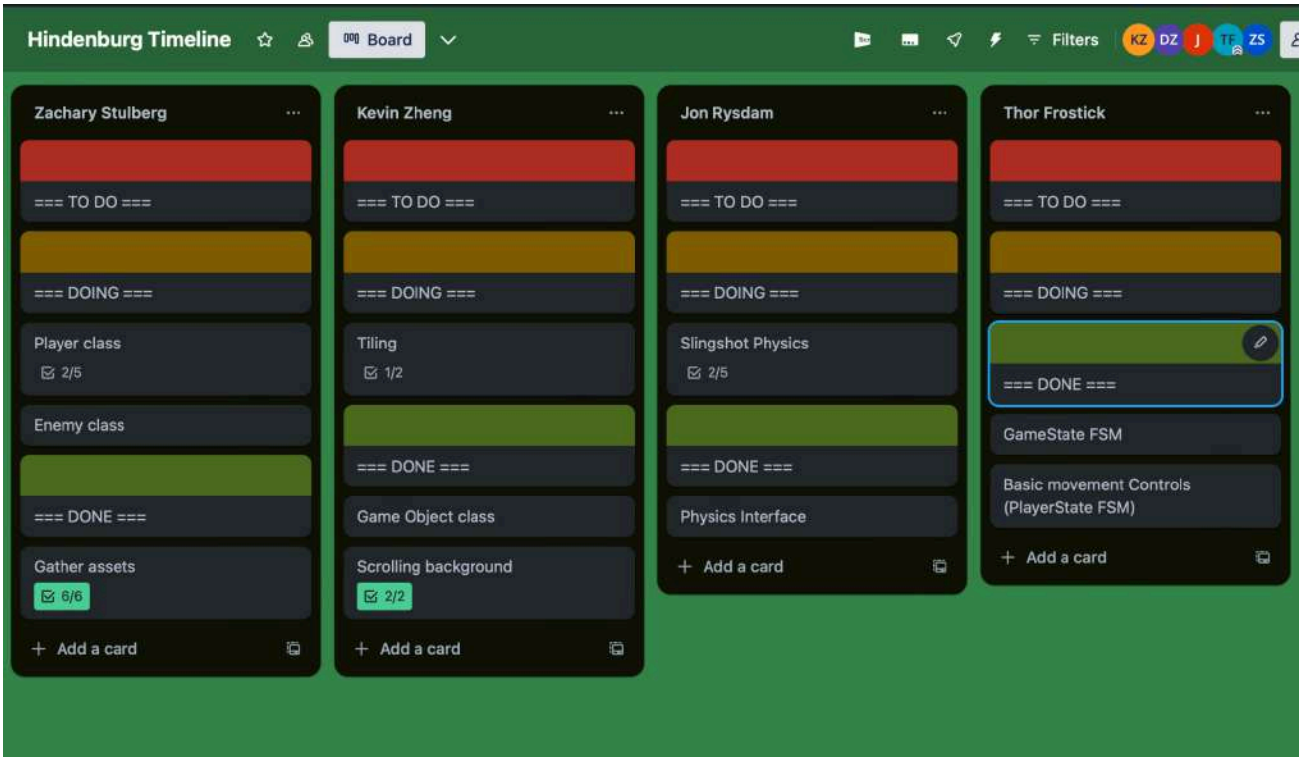
If the user interface has changed, include new mockups here:

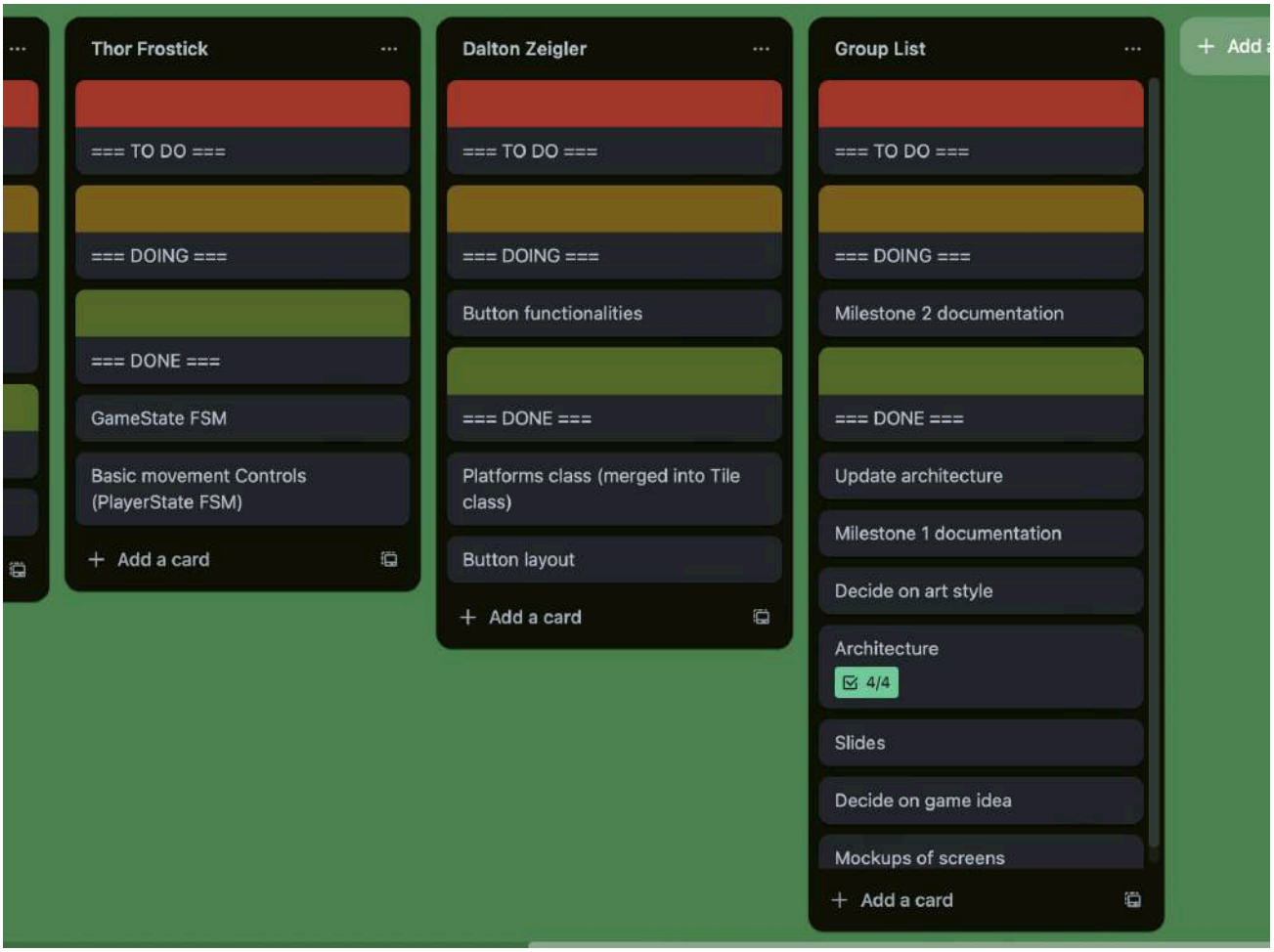
No changes.

If your method of task management has changed, include information about it here:

No changes.

Include a screenshot(s) of your Kanban board by the end of this milestone:





Task Timeline for Milestone 3

Include a bulleted list of tasks to complete during milestone 3:

- Fix the tiling and button functionality logic errors
- Timer in game
- Enemy class
- Implement assets/UI
- “Best time” board (similar to a leaderboard)

Which elements of your game are currently stretch goals?

More levels, collectibles, animation, and sound effects.

Lessons Learned

Write a 2 - 3 paragraph summary of lessons learned (both good and bad!) during this milestone.
This should be done after a group discussion of your group processes.

In the beginning, we began coding fluently, GameObject, Button, and GameState finite state machines were quickly done and incorporated into the group coding project. In addition, we also have a prototype of the grappling mechanic, which still needs to be improved at the time. We also decided to replace the grappling mechanic with something else after spring break if it still did not work. We even came up with some backup plans (high jumps, slingshot, and infinite jumps). Over spring break, we finished the IPhysics interface for the new slingshot feature, ditched grappling, and successfully implemented every button in the game.

Unfortunately, the pace slowed following the spring break. We begin to encounter issues with the scrolling background. We attempted to search for answers online but were unsuccessful. Fortunately, we still have the tutoring center and our fantastic professor. Although the issue was resolved, we spent too much time on this one problem which shrunk the time to work on other tasks. In addition, we are still having some difficulty finding the correct assets for our theme because there were not many free resources for abandoned labs.

Describe how your experience this milestone will influence your plans (workflow, communication, task management, etc.) for the next milestone.
Again, done in tandem with your group. What will you keep doing well? What changes will your group make?

We will continue communicating via discord and remind ourselves of the tasks that need to be completed on the Trello board. When we run into problems, we should ask for assistance from our professor, tutors, and one another. And try to stay on track for our tasks.

Milestone 3: Working Prototype

Link to your group's presentation slides:
Hindenburg Milestone 3 Slideshow

Scope

Has your plan for a viable MVP changed? Why? Describe your game's MVP.
(This is NOT just a list of features. Describe the target player experience of your prototype!)

We want the player to be able to open the game and move through the level, using a combination of clicking to fly onto new tiles and WASD to move on those tiles, all the while avoiding enemies in order to reach the top of the level, where a score and rank is given and saved to a leaderboard the player can challenge.

Team Responsibilities

List each team member's responsibilities during this milestone.

Team Member:	Responsibilities:
Thor Frostick	Leaderboard, End Conditions, Timer, Scores, Enemy
Jon Rysdam	Collisions
Zach Stulberg	Finishing Player, Enemy classes, Collisions
Kevin Zheng	Audio, Collisions, Background Tiles, God mode
Dalton Zeigler	GUI, Scores, Buttons

Production

How has your group utilized the task management strategy(ies) that you chose in the first milestone?

We have continued to use our Trello board to determine what tasks need to be completed, our Discord server to communicate with each other on areas we have worked on and various aspects of the game, as well as meeting regularly to work on the game.

Challenges

What challenges has your team faced with the project? How has your team overcome those challenges?

We had some issues getting the player to move properly and the tiles to properly recognize collision, but we managed to solve it through cooperative work and meeting together to solve the issues with all of our input combined.

Are there bugs that require fixing? If so:
What are they?
What is your team's plan for tracking and then fixing the bugs?

We have encountered some challenges with proper collision detection and switching between clicking and WASD controls.

Design/Architecture/Style/UI/Production Changes

If your design has changed, describe the changes here:

No changes.

If your architecture has changed, include an updated architecture diagram here:

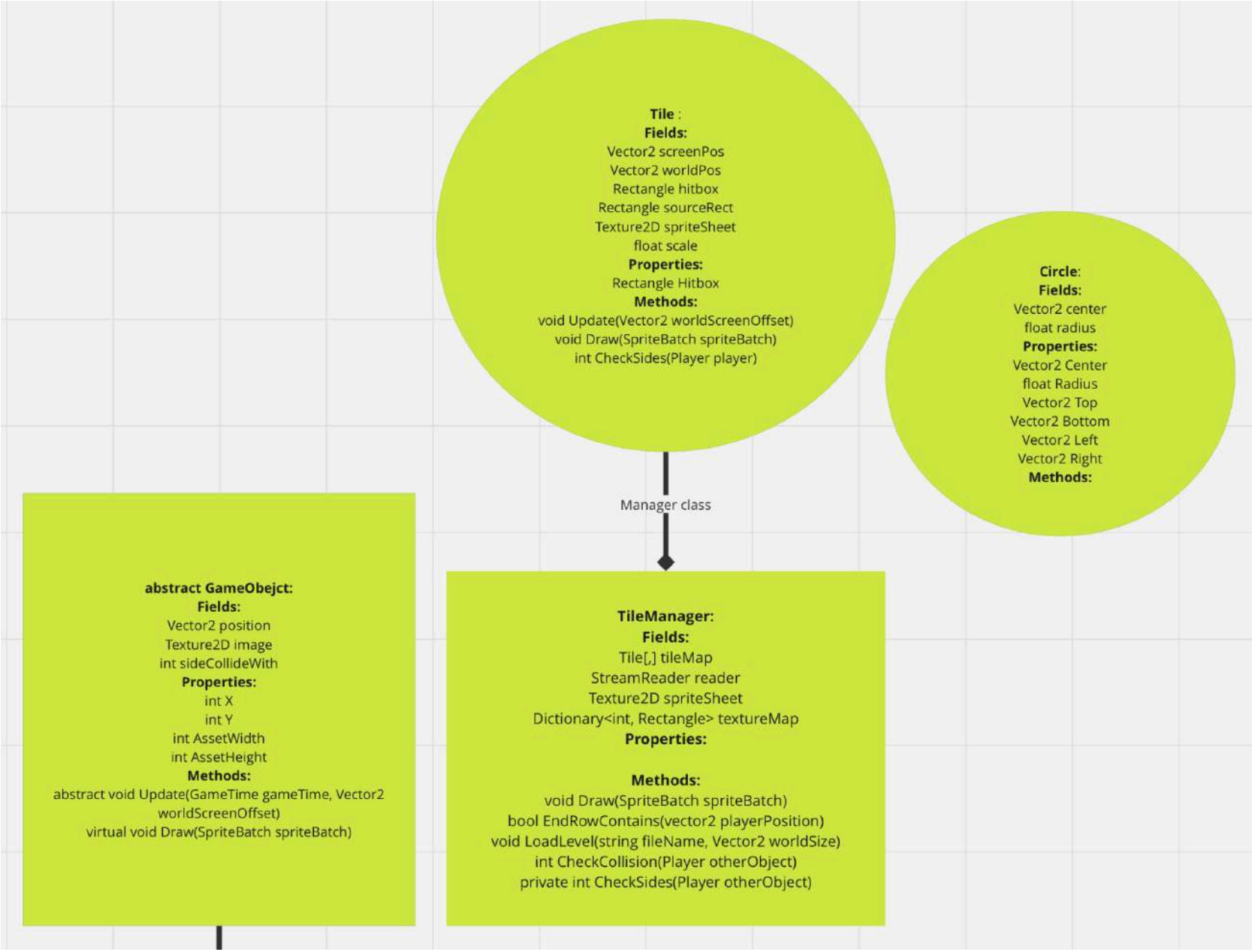
Game1:

Fields:

GameStates gameStates
GameStates previousGameState
bool gameFinished
bool godMode
TileManager tileManager
Texture2D tileSpriteSheet
Texture2D playerImg
Player slime
float scale
float speed
Texture enemyImg
Enemy spikeBall
float enemyScale
Vector2 playerWorldCenter
Vector2 worldSize
Vector2 playerWorldCenter
Background background
Vector2 screenSize
SpriteFont arial50
SpriteFont arial60
SpriteFont slimeFont120
SpriteFont buttonFont
List<Button> menuButtons
List<Button> pauseButtons
List<Button> gameOverButtons
List<Button> goBackButton
KeyboardState prevKBState
KeyboardState curKBState
double timer
double finalTime
string rank

Helper Methods:

bool SingleKeyPress(Keys key)
void StartGame()
void ResumeGame()
void Instructions()
void Back()
void StartAgain()
void Quit()


Tile :
Fields:
Vector2 screenPos
Vector2 worldPos
Rectangle hitbox
Rectangle sourceRect
Texture2D spriteSheet
float scale
Properties:
Rectangle Hitbox
Methods:
void Update(Vector2 worldScreenOffset)
void Draw(SpriteBatch spriteBatch)
int CheckSides(Player player)
Circle:
Fields:
Vector2 center
float radius
Properties:
Vector2 Center
float Radius
Vector2 Top
Vector2 Bottom
Vector2 Left
Vector2 Right
Methods:
Manager class
abstract GameObejct:
Fields:
Vector2 position
Texture2D image
int sideCollideWith
Properties:
int X
int Y
int AssetWidth
int AssetHeight
Methods:
abstract void Update(GameTime gameTime, Vector2 worldScreenOffset)
virtual void Draw(SpriteBatch spriteBatch)
TileManager:
Fields:
Tile[,] tileMap
StreamReader reader
Texture2D spriteSheet
Dictionary<int, Rectangle> textureMap
Properties:
Methods:
void Draw(SpriteBatch spriteBatch)
bool EndRowContains(vector2 playerPosition)
void LoadLevel(string fileName, Vector2 worldSize)
int CheckCollision(Player otherObject)
private int CheckSides(Player otherObject)

Player : GameObject, IPhysics

Fields:

PlayerState currentPlayerState
Vector2 worldPos
Vector2 velocity
Rectangle sourceRect
Circle hitbox
Circle nextHitbox
float scale
float speed
Vector2 screenSize
Vector2 worldSize
SoundEffectInstance hitSurface
SoundEffectInstance move
SoundEffectInstance slingshot

Properties:

Vector2 Position
Vector2 Velocity
int Width
int Height
int SideCollideWith
Circle Hitbox
Circle NextHitbox

Methods:

override void Update(GameTime gameTime)
void Draw(SpriteBatch spriteBatch, Vector2 worldScreenOffset)
void Reset(Vector2 startingPosition)
public void MovementUpdate()
public void SaveRank(double time, string fileName)
void LoadRank()
void Stun()

Background:
Fields:
Vector2 worldScreenOffset
Vector2 worldSize
TileManager backgroundTiles
Properties:
Vector2 WorldScreenOffset
TileManager BackgroundTiles
Methods:
void Update(Vector2 screenSize, Player player)
void Draw(SpriteBatch sb)

If your visual style has changed, describe the changes here and include new reference images:

No changes.

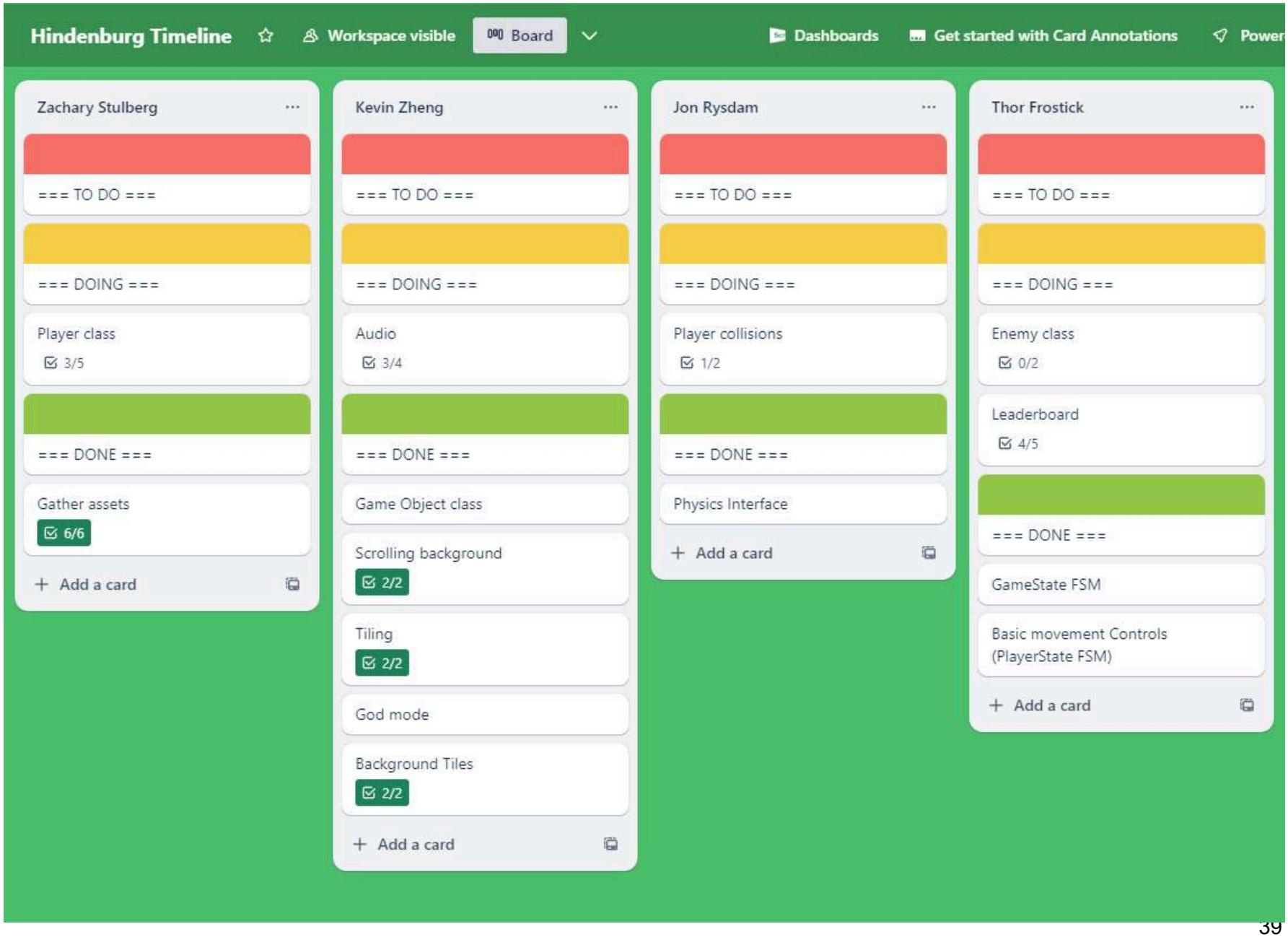
If the user interface has changed, include new mockups here:

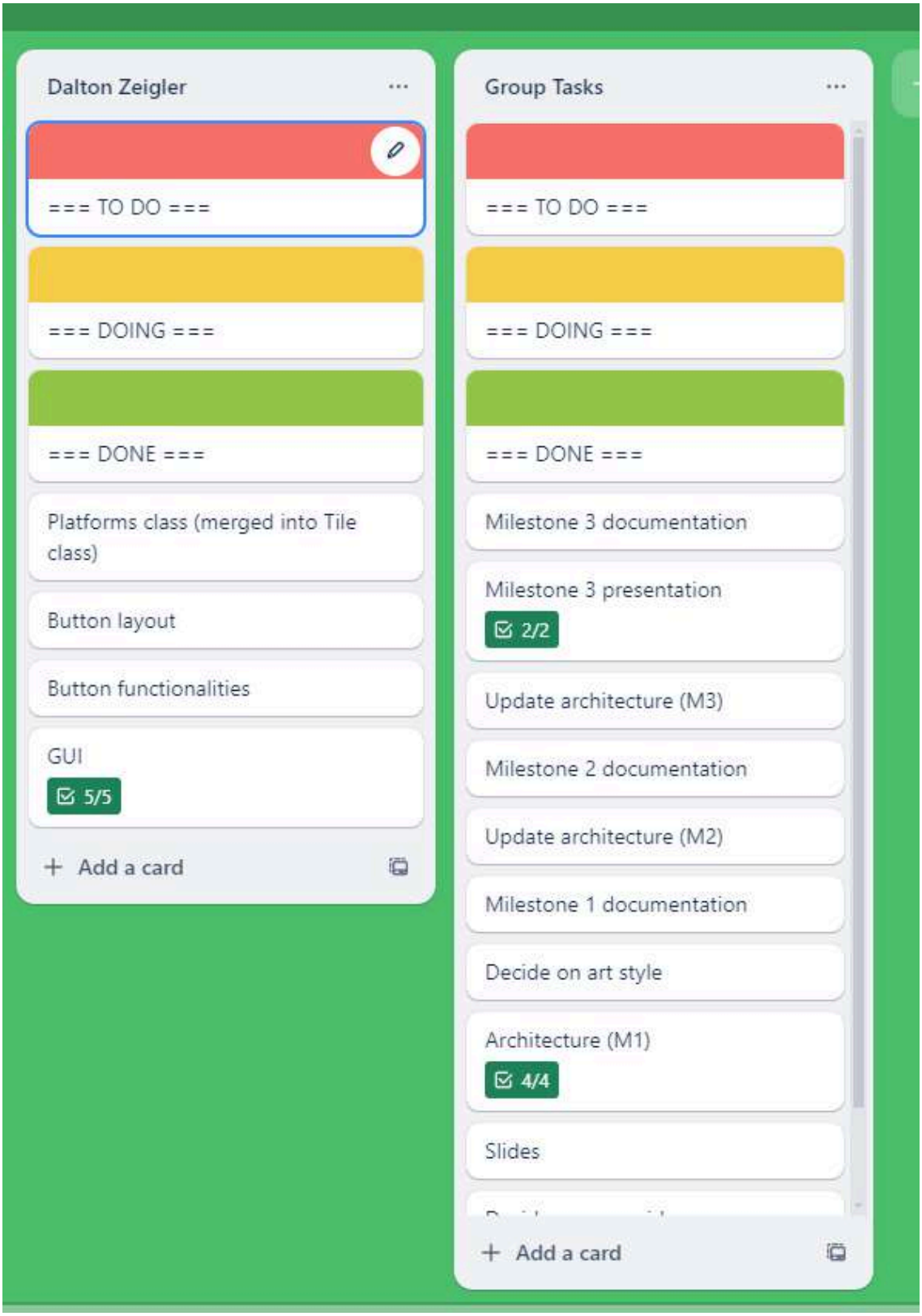
No changes.

If your method of task management has changed, include information about it here:

We have continued to use our Trello board and Discord server to discuss task management and what we will each be working on and completing.

Include a screenshot(s) of your Kanban board by the end of this milestone:


Hindenburg Timeline
Workspace visible
Board
Dashboards
Get started with Card Annotations
Power
Zachary Stulberg
=== TO DO ===
=== DOING ===
Player class
3/5
=== DONE ===
Gather assets
6/6
Add a card
Kevin Zheng
=== TO DO ===
=== DOING ===
Audio
3/4
=== DONE ===
Game Object class
Scrolling background
2/2
Tiling
2/2
God mode
Background Tiles
2/2
Add a card
Jon Rysdam
=== TO DO ===
=== DOING ===
Player collisions
1/2
=== DONE ===
Physics Interface
Add a card
Thor Frostick
=== TO DO ===
=== DOING ===
Enemy class
0/2
Leaderboard
4/5
=== DONE ===
GameState FSM
Basic movement Controls (PlayerState FSM)
Add a card


Dalton Zeigler
=== TO DO ===
=== DOING ===
=== DONE ===
Platforms class (merged into Tile class)
Button layout
Button functionalities
GUI
5/5
Add a card
Group Tasks
=== TO DO ===
=== DOING ===
=== DONE ===
Milestone 3 documentation
Milestone 3 presentation
2/2
Update architecture (M3)
Milestone 2 documentation
Update architecture (M2)
Milestone 1 documentation
Decide on art style
Architecture (M1)
4/4
Slides
Add a card

Task Timeline for Milestone 4

Include a bulleted list of tasks to complete during milestone 3:

- Complete implementing enemies and collisions between them and the player.
- Fix all issues surrounding collision with tiles and switching between WASD and mouse controls.
- Finish constructing a leaderboard that displays times and ranks in descending order from the top.
- Implement audio and sound effects.

Which elements of your game are currently stretch goals?

We are currently focused on polishing the level we have designed, though with our current file setup we may be able to create multiple levels to play. We have decided to keep the enemy objects staying on the tiles they spawn on, and if we have time we may make them fall off onto other tiles. Additionally, we will implement player and enemy animations only if we have enough time to do so.

Lessons Learned

Write a 2 - 3 paragraph summary of lessons learned (both good and bad!) during this milestone.
This should be done after a group discussion of your group processes.

At first, the project was being worked on at a steady pace, with prioritized tasks completed at a reasonable time. Communication was clear amongst the group members, including when members were going to start working on something and where and when meetings were going to take place. Basically, the project was on track for being completed pretty early.

However, there was a lot of procrastination from all of the members of the group, and certain tasks that needed to be worked on for an extended period of time were left to a later date, which slightly interfered with the pace at which the project would be completed. We have spoken with each other and everyone has taken full accountability of this fact. Moving forward for the next couple of weeks, we will be working hard to polish up our prototype, so that it is in a functional and organized state for the final deadline.

Describe how your experience this milestone will influence your plans (workflow, communication, task management, etc.) for the next milestone.
Again, done in tandem with your group. What will you keep doing well? What changes will your group make?

The procrastination experienced in this milestone was disadvantageous to the completion of the project. What could have been certain tasks spread out over a longer period of time has developed into these tasks needing to be completed over a smaller period of time. This means that each member must work longer/more frequently per week to make sure the project is completed come the next milestone. Therefore, with these past experiences and their consequences in mind, we have planned to communicate more often and meet with each other more frequently to work on getting this project finished.

Milestone 4: Finished Game

Link to your group's presentation slides:

Milestone 4 Slideshow

Team Responsibilities

List each team member's responsibilities during this milestone.

Team Member:	Responsibilities:
Thor Frostick	Enemy
Jon Rysdam	Collisions
Zach Stulberg	Enemy, Player
Kevin Zheng	Audio, Leaderboard
Dalton Zeigler	GUI, Buttons

Post Mortem

What went right?

Switching from one movement to another different movement

Leaderboard and Ranking

Buttons placement and functionality

Slingshot movement

Tiling and background scrolling

Efficiency in making and coding the game

What was not ideal?

Expanding screen size

Player collision
Enemy
Tiling, and buttons after expanding screen
Waiting last minute for to do documentation and slides
Asset collecting

What did your group learn from this process?

Dalton: How important communication is and how important to receive feedback, and always bounce ideas off of teammates to see if the idea is good or not, and ask advice or feedback on a method or code you created to see if it makes sense to have that method or code, or to make the code or method more efficient .

Jon: I have absolutely got to get a laptop that runs Windows, dual-boot or no. It is about 5 times harder to do anything without one. Also, all that stuff everyone else said about communication and commitment.

Thor : The process of making games is difficult. It requires heavy communication between all members of the team, flexibility in setting aside time to work on the project, and willingness to work through issues that may arise. Additionally, there are many issues that can arise during a development process that may not have simple solutions, and working alongside teammates is the only surefire way to find a solution that might work.

Kevin: Communication is the most important part of this process. Not only does that help to keep you and your teammates up to date with the code, you can also receive help/feedback from your teammates. Also, you cannot make everything yourself, this is why you have a team.

Zach: While it sounds cliché, making a functioning game is much more difficult than one would think. The work and code needed to be done is hard on its own, but alongside keeping communication between members and making sure certain tasks are completed within a certain time frame is also very difficult.

What will your group do differently next time?

Dalton: Make buttons easier to move, make buttons relay more on coordinates of screen, make the game full screen or make bigger borders of the game so player doesn't keep accidentally clicking of game,

Jon: Build the pieces of physics-related things step by step, and all positions should be relative to the world, not the screen. It is much simpler to only care about screen position when drawing.

Thor : I think it would be better for me to dedicate more time to the project, and perhaps work even closer with my teammates on solving bugs that arise. Additionally, I should put more focus and effort into completing the most difficult tasks when they become known, as I did not spend as much time and effort fixing issues surrounding the enemy objects. Moreover, I will make an effort to bring my charging cable to workdays, as I realized far too late I could work much more by extending my laptop's battery life.

Kevin: Utilize all my resources, for example, ask my teammates, tutors, or the professor for help as early as possible when there are problems so I can have more time to fix it or work on other things. In addition, try to remind others to do their work when the part that I am doing needs their code.

Zach: In my opinion, the three biggest things that I would do differently next time are better communication with the others in my group, less procrastination on getting tasks done, and spending more time working on the project itself.